CALIBRATION AND METROLOGY SERVICES (CAMS) V
STATEMENT OF WORK (SOW)

Overview

The NASA Johnson Space Center (JSC) in Houston, TX currently has an existing Measurement Standards and Calibration Laboratory (MSCL) located in Building 343.

The MSCL provides calibration operations for Measuring and Test Equipment (MTE) primarily for JSC Organizations. The MSCL also supports calibrations for other NASA centers and Federal Government Agencies. Most MTE are commercial off-the-shelf used to measure physical, mechanical, and electrical properties such as torque wrenches, pressure transducers and voltmeters.

CAMS V SOW requirements will be performed as Indefinite Delivery Indefinite Quantity (IDIQ) requirements, which will be issued as performance-based IDIQ Task Orders (TOs).

Deliverables required during performance of this SOW shall be provided by the Contractor per the Data Requirements Description (DRD) unless otherwise specified in the SOW or a TO.

Travel, training, and other direct costs for IDIQ requirements will be negotiated on a task order basis.

IDIQ requirements may change year-to-year and are negotiated each year and implemented as TOs. The Government may request, terminate or change IDIQ TOs at any time during the period of performance.

The structure of the SOW should not be construed as defining a required contractor organizational structure.

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1.0 CONTRACT MANAGEMENT

Acronyms

Acronyms utilized in this SOW are located in Appendix A – Acronyms.

Applicable Documents

The Contractor shall comply with the most current version of the following applicable documents:

- ANSI/ISO/ASQ (E) Q9001-2015, Quality Management Systems – Requirements
- ANSI/NCSL Z540-1-1994 (R2002), Calibration Laboratories and Measuring and Test Equipment – General Requirements
- ANSI/NCSL Z540.3-2006 (R2013), Requirements for the Calibration of Measuring and Test Equipment
- International Vocabulary of Metrology, Basic and General Concepts and Associated Terms (VIM) 3rd edition (JCGM 200:2012)
- ISO/IEC 17025:2017(E), General Requirements for the Competence of Testing and Calibration Laboratories
- JPD 8820.3, Facility Configuration Management (FCM) Program
- JPR 1281.11, Metrology and Calibration
- JPR 1281.14, Corrective and Preventive Actions and Continual Improvement
- JPR 1281.17, JSC Audits
- JPR 1440.3, JSC Records Management Procedural Requirements
- JPR 1700.1, JSC Safety and Health Requirements
- JPR 5322.1, Contamination Control Requirements Manual
- JPR 8550.1, JSC Environmental Compliance Procedural Requirements
- JPR 8553.1, Environmental Management System Manual
- JWI 1282.11, Calibration and Control of Measuring and Test Equipment (MTE)
- JWI 8553.1, Environmental Management System (EMS) Aspect/Impact Assessment and Environmental Management Plan (EMP) Process
- JWI 8831.1, Facility Manager Program
- MONRS Functional Specification dated 9/7/2011
- MONRS Technical Vision & Functional Requirements Rev 1.0 10/12/2011
- NPD 1440.6, NASA Records Management
- NPD 2810.1, NASA Information Security Policy
- NPR 1441.1, NASA Records Management Program Requirements
- NPR 2800.1, Managing Information Technology
- NPR 2810.1, Security of Information Technology
- NPR 8570.1, NASA Energy and Water Management Program
- SAE AS9100D, Quality Management Systems

These documents are available in the NASA Online Directives Information System (NODIS), the Center Directives Management System (CDMS), the JSC Quality Management System (QMS), or the MSCL Technical Library.

Management and Administration

The Contractor shall provide all data and perform all requirements of the Data Requirements Descriptions (DRDs), Attachment J-2, per the delivery schedule listed in the Data Requirements List, Attachment J-1.

The Contractor shall prepare and comply with a Contract Management Plan in accordance with DRD-08.

The Contractor shall ensure all personnel are qualified as defined in JPR 1281.11, Metrology and Calibration.

The Contractor shall provide a single point of contact and an alternate, to represent the Contractor on all contract issues and interface with the Contracting Officer (CO), Contracting Officer's Representative (COR) or their designees. The Contractor shall apprise the CO and COR immediately of any issues that could have an adverse impact on successful performance of the contract requirements.

The Contractor shall provide a Facility Manager for B-343 and participate in the JSC Facility Management Program (FMP) in accordance with JPD 8820.3, Facility Configuration Management (FCM) Program and JWI 8831.1, Facility Manager Program. The program helps to ensure a safe, healthy, and efficient workplace for all NASA JSC building occupants and provides for centralized reporting of facility issues.

2.0 PHASE-IN

The Contractor shall prepare and comply with a Contract Phase-in Plan in accordance with DRD-07. During phase-in, the Contractor shall work with the incumbent to ensure that the Contractor is prepared to fully execute the contract on the contract start date.

3.0 DOCUMENTATION AND RECORDS MANAGEMENT

The Government will provide existing documentation of the calibration laboratory procedures, technical manuals, equipment manuals, reference documents, reference standards reports, personnel training certifications, and calibration data packages. The Contractor shall maintain all documentation of the calibration laboratory procedures, technical manuals, equipment manuals, reference documents, reference standards reports, personnel training certifications, and calibration data packages required to meet the requirements of this SOW in the MSCL Technical Resource Library in accordance with ISO/IEC 17025:2017, ANSI/NCSL Z540-1-1994, and ANSI/NCSL Z540.3-2006.

The Contractor shall maintain records in accordance with the following Agency policies:

- JPR 1440.3, JSC Records Management Procedural Requirements
- JWI 1282.11, Calibration and Control of Measuring and Test Equipment (MTE)

- NPD 1440.6, NASA Records Management
- NPR 1441.1, NASA Records Management Program Requirements

The Contractor shall provide the Government or authorized representatives access to all records. The Government reserves the right to inspect, audit, and copy all records. The Contractor shall deliver all data in a format compatible with the current version of Microsoft Office Suite available on the Government-supplied desktops in the MSCL.

4.0 MSCL EQUIPMENT AND FACILITIES

The Government will supply equipment and facilities, including the calibration laboratory standards in the MSCL, in accordance with contract clauses NFS 1852.245-71 and Attachment J-3. Calibration laboratory standards are controlled reference devices that are used by the MSCL to verify the performance of less accurate devices.

The Government will provide to the Contractor desktop computers, telecommunications, network connectivity, and related services required in the performance of services covered by this Contract.

The Contractor shall maintain the cleanliness of the technical areas within the MSCL in accordance with JPD 8820.3, FCM Program. The Government is responsible for janitorial and grounds-keeping services.

The Contractor shall perform preventative maintenance on the MSCL standards to ensure operational condition per the equipment manuals.

5.0 INFORMATION TECHNOLOGY (IT)

The Contractor shall prepare and comply with an IT Security Management Plan in accordance with DRD-10.

The Contractor shall comply with NPR 2800.1, Managing Information Technology and NPR 2810.1, Security of Information Technology. The Contractor shall ensure that its employees, in performance of the contract, complete annual NASA IT security training by the designated due date. The Contractor shall use the System for Administration, Training, and Educational Resources for NASA (SATERN) or the current NASA training system to meet this requirement.

The Contractor shall obtain approval of all MSCL software changes by the appropriate Government Configuration Control Board (CCB) prior to implementation.

The Contractor shall ensure all software installed and maintained in the MSCL is compliant with NPD 2810.1, NASA Information Security Policy and NPR 2810.1, Security of Information Technology.

6.0 LOGISTICS

All calibration operations performed by the Contractor shall be performed either at the MSCL in B-343, at other JSC onsite locations, at Ellington Field, or at Sonny Carter Training Facility.

Calibration operations performed for MTE provided by customers at NASA Centers other than JSC or other Federal Agencies shall require the MTE to be shipped to and from JSC, at customer expense, and those calibration services shall be performed at the MSCL.

The Contractor shall use the JSC Shipping department located in building 420 for all items being shipped in support of this contract.

The Contractor shall coordinate, track, and document the pickup and delivery of all equipment leaving or entering the MSCL including commercial shipments sent through the JSC Shipping department.

The Contractor shall record the origin of items being received and the destination of items being shipped along with any other data needed to fully track the equipment such as carrier-provided tracking codes, purchase order information or Return Material Authorization (RMA) Codes.

The Contractor shall offer to provide pickup and delivery services for MTE from locations at or near JSC after issuance of a recall notice. The Contractor shall schedule these services upon customer request. Pickup and delivery shall be limited to JSC, Ellington Field, Sonny Carter Training Facility, and contractor facilities within a 10-mile radius of JSC. The Contractor shall provide the number and type of motor vehicles required to support the performance of this SOW in accordance with Attachment J-12.

7.0 SAFETY, HEALTH, AND ENVIRONMENTAL

7.1 Safety and Health

The Contractor shall ensure all their employees are knowledgeable of, and comply with, applicable federal, NASA, and JSC safety and health requirements, to include JPR 1700.1, JSC Safety and Health Requirements. JPR 1700.1 can be viewed at http://jschandbook.jsc.nasa.gov.

The Contractor shall ensure the protection of personnel, property, equipment, and the environment in all Contractor products and activities in this contract. To ensure compliance with pertinent NASA policies, requirements, federal, state, and local regulations for safety, health, environmental protection, and fire protection, the contractor shall develop and implement a safety and health program per a NASA-approved safety and health plan as described in DRD-03, Safety and Health Plan.

7.2 Quality Assurance and Reliability

The Contractor shall establish and maintain a QMS that is compliant with ANSI/ISO/ASQ Q9001-2015, Quality Management Systems and SAE AS9100D, Quality Management Systems, and in accordance with DRD-05, Quality Plan.

The Contractor shall maintain the MSCL's ISO/IEC 17025:2017, ANSI/NCSL Z540-1-1994, and ANSI/NCSL Z540.3-2006 accreditations.

The Contractor shall support NASA's internal and external audits and investigations of the MSCL performed by Government agencies or other authorized entities per JPR 1281.17, JSC Audits and JPR 1281.14, Corrective and Preventive Actions and Continual Improvement. This support may include, but is not limited to, responding to questions, attending meetings, providing data and documentation, and performing other related actions in response to requests from the auditors or investigators, in coordination with JSC/NT, Quality and Flight Equipment Division. The QMS established by the Contractor shall incorporate the following periodic product sampling methods for all MTE the MSCL is responsible for:

- Quality Review (QR): A complete review of MTE that has been calibrated prior to return to the customer. The MTE shall be randomly selected from production. The review shall include compliance to full parameter verification, physical condition, documentation, and traceability.

- Standard Review (SR): A complete review of a working standard used within the MSCL to calibrate other MTE. Standards shall be randomly selected from the production and the review shall include compliance to full parameter verification, physical condition, documentation, and traceability.

- Process Review (PR): A complete review of a documented calibration process to seek out and document procedural problems in order to correct them. Contractor management personnel shall observe the selected process as it normally occurs and document any process improvement opportunities and take the necessary corrective or preventative actions.

The QR and SR sampling rates shall be reviewed periodically in accordance with DRD-05.

The Contractor shall adjust sampling rates when negative or positive quality trends are observed, when changes occur in the overall skill level of the workforce, or when there are changes in the MSCL workload.

7.3 Environmental and Energy Conservation Requirements

The Contractor shall ensure that all work performed and equipment used to fulfill the requirements of this contract is in compliance with contract clause JSC 52.223-94, Environmental and Energy Conservation Requirements and Hazardous Materials Use.

8.0 CALIBRATION REQUIREMENTS

8.1 Calibration

The Contractor shall support the MSCL's normal business hours of 7:00 AM to 4:30 PM, Monday through Friday, except Federal Holidays. The MSCL shall remain open and operational on JSC's Quiet/Flex Fridays.

The contractor shall perform calibrations for MTE for JSC locations and from NASA centers other than JSC or Federal Government agencies, as directed by task order. The Contractor's calibrations performed for non-JSC organizations shall meet all standards and requirements of this contract, with any exceptions documented in the individual task orders issued.

The Contractor shall maintain metrology reference, transfer, and working standards in accordance with ISO/IEC 17025:2017, ANSI/NCSL Z540-1-1994 and ANSI/NCSL Z540.3-2006 accreditations. The calibration standards used at JSC shall be traceable to the National Institute of Standards and Technology (NIST). The Contractor shall maintain and operate the MSCL standards and capabilities according to the ranges and parameters specified in Attachment J-5, Summary of MSCL Calibration Disciplines and Ranges.

The Contractor shall prepare and archive test reports documenting testing operations performed on reference, transfer, and working standards. The results of these operations shall be recorded in the Metrology Information Management System (MIMS) database. Test reports shall contain a description of the test, the test conditions, a tabulation of results, statement of any limitation in use of the unit under test, with the appropriate signature attesting to the authenticity of such test.

The Contractor shall perform calibration operations in accordance with JPR 1281.11 and JWI 1282.11. The Contractor shall perform calibrations of the active MTE identified in MIMS. Active MTE are those MTE currently scheduled to be calibrated including in-situ and outsourced MTE. Approximately 1,400 MTE migrate between the active and inactive instrument lists per year.

The Contractor shall be responsible for the approval or disapproval, authorization, and processing of all outsourced calibration services per JPR 1281.11 and JWI 1282.11.

The Contractor shall qualify and maintain a listing of approved calibration service providers (third party calibration vendors) for MTE that cannot be serviced by the MSCL in accordance with JPR 1281.11 and JWI 1282.11. The Contractor shall evaluate the outside suppliers to ensure compliance with ISO/IEC 17025:2017, ANSI/NCSL Z540-1-1994, and ANSI/NCSL Z540.3-2006. The Contractor shall record all calibration data provided by outside suppliers in the MIMS database. The Contractor shall assign re-calibration dates for and apply labels to equipment calibrated by outside suppliers in accordance with JWI 1282.11.

8.2 Software Tools and Calibration Data Management

The Contractor shall update and maintain the software for the MSCL's Metrology Information Management System (MIMS), which includes the MET/CAL calibration management software suite, along with the Metrology Out-of-Tolerance Notification Reporting System (MONRS) software tool. The MET/CAL software suite houses the MTE Inventory database, and includes the MET/CAL procedure editor that is used to support performance of the actual calibrations. MET/CAL also includes the MET/TEAM software, that is used for calibration work flow and asset management. MONRS is a tool that was developed in-house at JSC for tracking out-of-tolerance calibrations. The Contractor shall manage user access and purchase software licenses for the MET/CAL software suite.

The Contractor shall update and maintain the records in the MTE Inventory database in MIMS, in accordance with JPR 1281.11, Metrology and Calibration. MIMS contains data such as MTE calibration and repair records, calibration standard measurement uncertainty determinations, software control records, and other calibration records required to operate the calibration lab and maintain compliance with JPR 1281.11.

The Contractor shall update and maintain the data within the MONRS software tool in accordance with JWI 1282.11, Calibration and Control of Measuring and Test Equipment (MTE), the MONRS Technical Vision & Functional Requirements Rev 1.0 dated 10/12/2011, and the MONRS Functional Specification dated 9/7/2011.

The Contractor shall initiate issuance of an electronic notification in MONRS to the MTE user when an Out-of-Tolerance (OOT) condition is discovered.

The Contractor shall monitor the progress of the MTE user to ensure completion and disposition of the non-conforming condition of the MTE, and the completion of the appropriate selections as directed by the MONRS tool and in accordance with JPR 1281.11 and JWI 1282.11.

The Contractor shall provide the COR with status of open actions on OOT conditions as required in accordance with DRD-06, Performance Metrics Reports.

8.3 Calibration Turn-Around Time and Commit Dates

The Contractor shall complete 85% of the MTE calibrations within a 10-day turn-around time. Turn-around time is calculated from the day the item is received in the MSCL to the day the customer has been notified the calibration is complete and only includes business days. The standards included in Attachment J-3, Installation-Accountable Government Property (IAGP), are exempt from turnaround time reporting. Exceptions include calibrations that are performed by outside suppliers or in an authorized hold status listed below:

a. Customer Hold (waiting customer response)
b. Lab Hold (awaiting information from Original Equipment Manufacturer)
c. Procedure Hold (item is awaiting completion of procedure documentation update)
d. Repair Parts Hold (waiting on repair parts)

The Contractor shall submit recall notices to customers in accordance with JWI 1282.11.

The Contractor shall accept a maximum of 2% of any customer's inventory on priority basis per contract year. A priority calibration is one that is requested by a customer prior to the next scheduled servicing time for the specific MTE. The Contractor shall perform priority calibration requests in accordance with the procedures documented in JWI 1282.11, Calibration and Control of Measuring and Test Equipment (MTE).

8.4 Technical Representation

The Contractor shall supply technical representation for meetings with NASA, JSC customers, and JSC Contractors as part of normal business operations. These meetings include:

- COR Status Meeting – Weekly
- NASA Metrology and Calibration Working Group (MCWG) – Weekly video/teleconference
- Scheduling meetings – Bi-Weekly
- Technical discussions – Bi-Weekly
- Control boards – Bi-Annually
- Contractor Status Report – Annual per DRD-04, Calibration and Metrology Periodic Progress Reports
- NASA MCWG Meeting – Annual; rotates between NASA Centers; participation requires the Contractor to prepare and give presentations on the technical aspects of the MSCL

8.5 Metrology

The Contractor shall participate in the NASA Measurement Assurance Program (MAP) by performing comparison measurements between the MSCL's reference standards and MAP standards sent to the MSCL by other NASA centers. The NASA MAP is a collaboration among calibration labs at various NASA Centers. Each calibration lab maintains reference standards in different areas of specialization such as voltage, capacitance or mass. These standards are rotated among the NASA Centers and are compared to their reference standards to ensure the quality of measurements made within NASA programs and to establish measurement uncertainties.

8.6 REPAIR AND REPLACE

The contractor shall repair and replace the MSCL standards and other MSCL equipment as directed by task order.

Appendix A
Acronyms

Calibration and Metrology Services (CAMS)
Center Directives Management System (CDMS)
Commercial-Off-The-Shelf (COTS)
Configuration Control Board (CCB)
Contracting Officer (CO)
Contracting Officer's Representative (COR)
Data Requirements Description (DRD)
Environmental Management Plan (EMP)
Environmental Management System (EMS)
Facility Configuration Management (FCM)
Facility Management Program (FMP)
Indefinite Delivery Indefinite Quantity (IDIQ)
Information Technology (IT)
Installation-Accountable Government Property (IAGP)
Johnson Space Center (JSC)
Measurement Assurance Program (MAP)
Measuring and Test Equipment (MTE)
Measurement Standards and Calibration Laboratory (MSCL)
Metrology and Calibration Working Group (MCWG)
Metrology Information Management System (MIMS)
Metrology Out-of-Tolerance Notification Reporting System (MONRS)
National Aeronautics and Space Administration (NASA)
NASA Online Directives Information System (NODIS)
National Institute of Standards and Technology (NIST)
Out-of-Tolerance (OOT)
Process Review (PR)
Quality Management System (QMS)
Quality Review (QR)
Return Material Authorization (RMA)
Standard Review (SR)
Statement of Work (SOW)
System for Administration, Training, and Educational Resources for NASA (SATERN)
Technical Management Representative (TMR)

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